The NOSS Classification System: A 12-Sector Taxonomy for Computational Cognitive Assessment

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The author is the founder and president of Beverly Index LLC, a company developing commercial applications based on the classification system presented in this paper. This potential conflict of interest is disclosed in accordance with APA ethical standards. The classification system presented herein is offered for scientific scrutiny and empirical testing by the research community.

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Abstract

This paper presents the Neural Oscillatory Spectroscopy System (NOSS), a 12-sector classification taxonomy designed to operationalize Neural Oscillation Signature Theory (NOST; Beverly, 2026b) for computational cognitive assessment. NOSS organizes the space of clinically relevant cognitive states into 12 sectors, each defined by its primary oscillatory disruption profile, the cognitive processes those disruptions are expected to affect, and the linguistic features those cognitive changes are predicted to produce. The 12 sectors encompass mood regulation, anxiety and threat-response, psychotic-spectrum processing, neurocognitive function, developmental processing differences, optimal cognitive baseline, traumatic brain injury, neurodegenerative conditions, captivity and coercion trauma, substance-related cognitive distortion, dissociative processing, and personality-related patterns. NOSS addresses a critical gap in computational approaches to mental health assessment by providing a theoretically grounded classification framework that maps the complete clinical landscape to predicted oscillatory–linguistic signatures. The system employs multi-label, continuous-valued activation rather than categorical diagnosis, acknowledging that cognitive states rarely conform to single diagnostic categories. This paper specifies each sector's theoretical basis, predicted oscillatory signature, expected linguistic markers, clinical conditions encompassed, and cross-sector interaction dynamics. NOSS is offered as a theoretical classification system warranting empirical validation.

Keywords: classification system, cognitive assessment, neural oscillations, linguistic markers, NOSS, computational psychiatry, taxonomy, multi-label classification

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Neural Oscillation Signature Theory (NOST; Beverly, 2026b) proposes that cognitive states leave detectable signatures in linguistic output through the mediating influence of neural oscillatory processes on language production. If NOST's central claim is correct—that the statistical properties of language carry extractable information about the oscillatory configuration that generated them—then a systematic classification system is needed to organize the space of cognitive states that the framework aims to assess.

This paper introduces such a system: the Neural Oscillatory Spectroscopy System (NOSS). The name reflects the system's conceptual analogy to spectroscopy in physics and chemistry—the decomposition of a complex signal into its constituent frequency components to identify the source. Just as optical spectroscopy decomposes light into wavelengths to identify chemical elements, NOSS proposes to decompose linguistic output into oscillatory-band-associated feature classes to identify the cognitive states that generated it.

NOSS is designed to serve as the operational bridge between NOST's theoretical claims and their clinical application. It translates the abstract proposition that oscillatory states produce linguistic signatures into a concrete taxonomy of 12 sectors, each with specified predicted signatures that can be empirically validated or refuted.

Design Principles

Three principles guided the construction of NOSS.

Principle 1: Multi-Label Continuous Activation

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Human cognitive states rarely conform to single diagnostic categories. Comorbidity is the rule rather than the exception in psychiatric diagnosis (Kessler et al., 2005). NOSS therefore employs a multi-label, continuous-valued activation model: each sector receives an activation score on a continuous scale, and multiple sectors may be simultaneously elevated. The output of a NOSS assessment is a 12-dimensional activation vector, not a categorical label. This design aligns with dimensional approaches to psychopathology (Kotov et al., 2017) while preserving clinical utility through interpretable sector-level scores.

Principle 2: Oscillatory Grounding

Each sector is defined not merely by its clinical phenomenology but by its predicted oscillatory disruption profile—the pattern of neural oscillatory changes that NOST predicts should underlie the cognitive state and produce its characteristic linguistic signature. This grounding in oscillatory neuroscience provides mechanistic specificity that purely phenomenological classification systems lack and generates predictions that can be tested with concurrent EEG–language measurement.

Principle 3: Clinical Comprehensiveness

NOSS aims to cover the major domains of clinical cognitive assessment. The 12 sectors were selected to encompass the conditions most frequently encountered in clinical practice while also including domains (such as captivity trauma and neurodegenerative conditions) that have been inadequately addressed by existing assessment frameworks. The system is designed to be extensible: additional sectors can be added as the theoretical and empirical base expands.

The 12-Sector Classification

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Table 1 provides an overview of all 12 NOSS sectors. The following sections develop each sector in detail.

Table 1

The 12 NOSS Sectors: Overview

Code	Sector	Clinical Domain
NOSS-S1	Mood	Affective regulation, emotional baseline stability, bipolar cycling
NOSS-S2	Anxiety	Threat-response, hypervigilance, avoidance, PTSD, autonomic arousal
NOSS-S3	Psychotic	Reality-processing integrity, thought organization, perceptual accuracy
NOSS-S4	Neurocognitive	Executive function, memory, attention, processing speed
NOSS-S5	Developmental	ADHD, ASD, neurodevelopmental processing differences
NOSS-S6	Optimal	Healthy cognitive baseline, resilience, cognitive reserve
NOSS-S7	TBI	Traumatic brain injury, concussion, post-concussive syndrome
NOSS-S8	Neurodegenerative	Parkinson's disease, CTE, progressive cognitive-motor decline
NOSS-S9	Captivity Trauma	Coercion, trafficking, learned helplessness, identity suppression
NOSS-S10	Substance	Addiction, reward distortion, craving signatures
NOSS-S11	Dissociative	Fragmentation, depersonalization, derealization
NOSS-S12	Personality	Enduring interpersonal patterns, attachment, identity stability

Sector S1: Mood

The Mood sector encompasses conditions characterized by disruptions to affective regulation and emotional baseline stability, including major depressive disorder, bipolar spectrum disorders, persistent depressive disorder, and premenstrual dysphoric disorder. The predicted oscillatory profile involves frontal alpha asymmetry (reduced left frontal alpha power in depressive states; Thibodeau et al., 2006), disrupted reward-related theta-band activity in the ventral striatum (Knyazev, 2007), and altered gamma synchrony during emotional processing.

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Predicted linguistic markers include increased first-person singular pronoun use, elevated negative emotion vocabulary, reduced future tense reference, absolutist language patterns (Al-Mosaiwi & Johnstone, 2018), diminished syntactic complexity during depressive episodes, and pressured speech with increased word output and semantic tangentiality during manic episodes. The sector further predicts that bipolar cycling should produce measurable oscillation in these linguistic features across mood states, providing a longitudinal signature distinct from unipolar depression.

Sector S2: Anxiety

The Anxiety sector encompasses conditions characterized by disrupted threat-response processing, including generalized anxiety disorder, panic disorder, social anxiety disorder, post-traumatic stress disorder, and obsessive-compulsive spectrum conditions. The predicted oscillatory profile involves heightened beta-band activity associated with hypervigilance, disrupted alpha inhibition of threat-irrelevant processing, and altered theta-band activity in hippocampal-amygdalar circuits affecting fear memory consolidation (Lesting et al., 2011).

Predicted linguistic markers include elevated threat-related vocabulary, increased hedging and uncertainty markers, avoidant language surrounding specific content domains, temporal fragmentation in trauma narratives, reduced narrative integration of threatening material, and increased use of safety-seeking and reassurance-oriented language. The sector distinguishes between generalized anxiety (diffuse linguistic hypervigilance) and trauma-specific anxiety (content-specific avoidance with otherwise intact language) as distinct signature subtypes.

Sector S3: Psychotic

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The Psychotic sector encompasses conditions characterized by disruptions to reality-processing integrity, including schizophrenia spectrum disorders, schizoaffective disorder, and psychotic features in other conditions. The predicted oscillatory profile involves the well-documented reduction in gamma-band synchrony (Uhlhaas & Singer, 2010), disrupted theta-gamma coupling, and aberrant beta-band activity associated with distorted predictive processing.

Predicted linguistic markers include reduced semantic coherence across discourse segments, increased referential ambiguity, tangential and derailed discourse, loosened associations, neologisms, and disrupted pragmatic language use. These predictions are among the best supported by existing empirical evidence (Bedi et al., 2015; Elvevåg et al., 2007; Corcoran et al., 2018), lending credibility to NOSS's broader classification approach.

Sector S4: Neurocognitive

The Neurocognitive sector encompasses conditions characterized by primary deficits in executive function, memory, attention, and processing speed, including mild cognitive impairment, Alzheimer's disease (early stages), vascular cognitive impairment, and age-related cognitive decline. The predicted oscillatory profile involves reduced theta-band power during memory tasks, diminished alpha desynchronization during attentional engagement, and slowed dominant frequency reflecting reduced processing speed.

Predicted linguistic markers include progressive reduction in vocabulary diversity (type-token ratio), simplification of syntactic structures, increased word-finding pauses and circumlocutions, reduced propositional density, and diminished temporal coherence in narrative construction. These predictions align with the existing literature on linguistic markers of cognitive decline (Fraser et al., 2016; Garrard et al., 2005) and provide the basis for early

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detection of neurocognitive changes before they reach clinical thresholds on standard assessment instruments.

Sector S5: Developmental

The Developmental sector encompasses neurodevelopmental conditions including attention-deficit/hyperactivity disorder, autism spectrum conditions, and specific learning differences. The predicted oscillatory profile for ADHD involves elevated theta-to-beta ratio (Arns et al., 2013) and reduced frontal beta-band activity during sustained attention. For autism spectrum conditions, the profile involves altered gamma-band activity during social processing and atypical cross-frequency coupling patterns.

Predicted linguistic markers for ADHD include increased topic shifts, reduced coherence maintenance across extended discourse, and elevated dysfluency rates during attentionally demanding tasks. For autism spectrum conditions, predicted markers include atypical prosodic patterns, reduced pragmatic language markers, literal interpretation patterns, and distinctive patterns of information organization in narrative construction. The sector acknowledges the significant heterogeneity within neurodevelopmental conditions and proposes that oscillatory–linguistic profiling may contribute to more precise characterization of individual profiles within these broad diagnostic categories.

Sector S6: Optimal

The Optimal sector serves as the reference baseline for the classification system, representing healthy cognitive function, emotional resilience, and cognitive reserve. The predicted oscillatory profile involves well-organized alpha rhythm with appropriate task-related desynchronization, efficient theta-mediated memory operations, appropriate beta-band

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modulation during cognitive control tasks, and robust gamma synchrony during perceptual and semantic processing.

The predicted linguistic signature of optimal function includes rich vocabulary diversity, complex and varied syntactic structures, coherent narrative organization with appropriate temporal reference, balanced emotional valence, and efficient information density. The Optimal sector is critical for calibrating the clinical sensitivity of the system: deviations from the optimal profile in specific sectors define the clinical activation pattern. The sector also provides the framework for characterizing cognitive reserve—the degree to which an individual's linguistic output reflects maintained cognitive sophistication despite advancing age or early pathological processes.

Sector S7: Traumatic Brain Injury

The TBI sector encompasses acute concussion, post-concussive syndrome, moderate-severe traumatic brain injury, chronic traumatic encephalopathy risk profiles, and repetitive head impact exposure. The predicted oscillatory profile involves elevated delta and theta power during wakefulness (indicating cortical slowing), disrupted cross-frequency coupling patterns, and reduced beta-band modulation during cognitive control tasks (Palacios et al., 2019; Rapp et al., 2015).

Predicted linguistic markers include reduced global narrative coherence, increased processing time (pauses, hesitations), simplified syntactic structures, diminished executive language functions (planning, sequencing in discourse), and specific patterns of attention-related errors. The TBI sector is designed to detect cognitive effects of head injury even when structural neuroimaging appears normal—a well-documented clinical scenario (Bigler, 2013)—by identifying the subtle linguistic consequences of oscillatory disruption.

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Sector S8: Neurodegenerative

The Neurodegenerative sector encompasses progressive conditions including Parkinson's disease, Huntington's disease, amyotrophic lateral sclerosis (cognitive components), and progressive supranuclear palsy. The predicted oscillatory profile for Parkinson's disease—the most thoroughly characterized condition in this sector—involves pathological beta-band oscillations in basal ganglia–cortical circuits (Little & Brown, 2014) and disrupted dopaminergic modulation of theta-band activity.

Predicted linguistic markers for Parkinson's disease include a distinctive combined motor-speech and cognitive-linguistic signature: reduced articulation rate, monotone prosody, vocal fold tremor at characteristic frequencies, reduced action verb diversity, simplified sequential planning in discourse, and diminished motor-related language complexity. The sector proposes that this combined signature may be detectable in prodromal stages, particularly in conjunction with REM sleep behavior disorder—a known prodromal marker of Parkinson's disease (Postuma et al., 2012)—that may produce distinctive dream narrative signatures. If validated, this would represent a significant advance in early detection of neurodegenerative conditions.

Sector S9: Captivity Trauma

The Captivity Trauma sector represents a novel addition to cognitive classification systems, encompassing the distinct psychological sequelae of prolonged coercive control, captivity, trafficking, and torture. This sector is distinguished from the general Anxiety sector (S2) by its focus on conditions involving sustained identity suppression, coerced behavioral compliance, and the complex adaptations that emerge under prolonged captivity—conditions that produce a psychological profile distinct from standard PTSD (Herman, 1992).

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The predicted oscillatory profile involves chronic hyperarousal patterns (sustained beta elevation), disrupted theta-band processing of autobiographical memory (reflecting identity fragmentation), and altered gamma synchrony during self-referential processing. Predicted linguistic markers include identity suppression indicators (reduced first-person agency, externalized locus of control), compliance-oriented language patterns, hypervigilance markers distinct from general anxiety, temporal disorientation in autobiographical narrative, and specific patterns of moral injury language reflecting internalized shame and guilt associated with survival behaviors.

The sector encompasses several sub-conditions: captivity-related complex PTSD, torture trauma syndrome, psychological torture and coercive control effects, Stockholm syndrome patterns, moral injury, and reintegration difficulty. Each sub-condition is predicted to produce distinguishable linguistic signature subtypes within the broader captivity trauma profile. This sector has particular relevance for military repatriation assessment, human trafficking identification, and forensic evaluation of coercion claims.

Sector S10: Substance

The Substance sector encompasses conditions related to substance use disorders, including active addiction, early recovery, and sustained recovery states. The predicted oscillatory profile involves disrupted theta-band reward processing in mesolimbic circuits, altered beta-band inhibitory control, and dysregulated alpha-band resting-state activity (Newson & Thiagarajan, 2019).

Predicted linguistic markers include reward-biased vocabulary, reduced temporal discounting markers (preference for immediate over delayed in narrative framing), craving-related language during periods of abstinence, diminished executive language complexity during

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active use, and specific patterns of denial, minimization, and externalization. The sector proposes that linguistic signatures should track recovery trajectory, with progressive normalization of executive language complexity and reward-related vocabulary bias during sustained recovery.

Sector S11: Dissociative

The Dissociative sector encompasses conditions characterized by disruptions to integrative psychological functions, including dissociative identity disorder, depersonalization/derealization disorder, dissociative amnesia, and dissociative features in other conditions. The predicted oscillatory profile involves disrupted cross-frequency coupling (reflecting failure of integrative processing), atypical alpha-band activity during self-referential processing, and altered gamma synchrony during perceptual integration.

Predicted linguistic markers include narrative fragmentation (abrupt shifts in perspective, tense, or register), depersonalization indicators (third-person self-reference, emotional detachment markers), gaps and discontinuities in autobiographical narrative, and distinctive patterns of sensory-perceptual description suggesting altered integration of perceptual experience. The Dissociative sector interacts significantly with the Captivity Trauma sector (S9), as dissociative processing frequently develops as an adaptive response to prolonged captivity and coercion.

Sector S12: Personality

The Personality sector encompasses enduring patterns of interpersonal functioning, identity, and self-regulation that are clinically significant, including personality disorder features across the spectrum described in the Alternative DSM-5 Model for Personality Disorders. The predicted oscillatory profile involves stable, trait-like patterns of oscillatory organization—in

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contrast to the state-dependent disruptions characterizing most other sectors—reflecting the enduring nature of personality-related processing.

Predicted linguistic markers include characteristic interpersonal language patterns (dominance, submission, affiliation, hostility markers), identity stability indicators (consistency of self-reference across contexts), emotional regulation patterns (volatility, suppression, or hyper-regulation), and distinctive attachment-related language in narratives involving relational themes. The Personality sector is unique within NOSS in that it seeks to characterize enduring processing styles rather than acute or episodic cognitive states, requiring longitudinal observation and cross-contextual consistency analysis.

Cross-Sector Dynamics

A critical feature of NOSS is its capacity to model comorbid and interacting cognitive states through simultaneous multi-sector activation. Three types of cross-sector dynamics are proposed.

Co-Activation

Many clinical presentations involve elevated activation in multiple sectors simultaneously. For example, a comorbid presentation of depression and anxiety would produce elevated activation in both S1 (Mood) and S2 (Anxiety), with the relative magnitudes and specific feature profiles reflecting the individual's unique clinical picture. NOSS's continuous activation model accommodates this naturally, avoiding the forced-choice limitations of categorical diagnosis.

Masking

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In some configurations, activation in one sector may suppress or mask the linguistic signature of another sector. For example, prominent psychotic features (S3) may obscure the linguistic markers of underlying mood pathology (S1). NOSS proposes that careful analysis of the full 12-dimensional activation vector, including the specific features contributing to each sector's activation, can disambiguate such cases.

Cascading

Some sector activations may produce secondary elevation in related sectors through known pathological cascading. For example, traumatic brain injury (S7) frequently leads to secondary mood disturbance (S1) and anxiety (S2). Substance use (S10) may produce neurocognitive effects (S4) or trigger psychotic features (S3). NOSS models these as expected cascading patterns that, when observed, provide additional diagnostic confidence.

Formal Specification of the Activation Model

The NOSS output is a 12-dimensional activation vector: $a = (a1, a2, ..., a12)$ where each component ai is an element of the continuous interval [0, 100] and corresponds to one of the 12 clinical sectors. Each sector is characterized by a theoretically predicted profile specifying the linguistic features expected to be elevated or diminished under the cognitive state characteristic of that sector.

Sector activation levels are derived from the correspondence between observed linguistic features and sector-specific theoretical predictions. The proprietary computational methodology underlying activation scoring is detailed in a separate technical monograph (Beverly, 2026d) and is not described here.

This continuous-valued multi-label approach represents a departure from traditional categorical diagnostic systems. The 12-dimensional activation vector provides a richer

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description of cognitive state than a single diagnostic label, captures the graded nature of most cognitive disruptions, and facilitates longitudinal tracking of changes across multiple dimensions simultaneously.

An important theoretical question is how a continuous activation model can nevertheless produce clinically meaningful threshold behavior—that is, how gradual changes in underlying oscillatory state can produce what clinicians observe as relatively sharp transitions between functional and clinically significant presentations. Recent findings in particle physics offer a relevant precedent. Studies of the quark-hadron phase transition at the Quark Matter 2025 conference demonstrated that the underlying chiral symmetry crossover in quantum chromodynamics is smooth and continuous, yet the observable transition from hadron resonance gas to quark-gluon plasma occurs abruptly at a well-defined temperature (Andronic et al., 2025). A smooth underlying change produces a sharp observable transition. NOSS proposes an analogous structure: the underlying oscillatory changes driving sector activation are continuous, but the linguistic and behavioral consequences of those changes may manifest as relatively sharp threshold crossings when the oscillatory configuration passes a critical point at which a qualitatively different cognitive processing mode is engaged. This observation suggests that NOSS sector activations, while measured on a continuous scale, may in practice exhibit nonlinear threshold dynamics—a prediction amenable to empirical testing through detailed analysis of the activation score distributions observed in clinical samples.

Validation Requirements

The empirical validation of NOSS requires several sequential steps. First, the predicted linguistic markers for each sector must be tested against clinical samples with established diagnoses. This requires large, well-characterized samples for each sector—a substantial data

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collection challenge, particularly for less common conditions. Second, the discrimination between sectors must be assessed: can the system distinguish between conditions with overlapping linguistic features (e.g., depression versus neurocognitive decline, both of which may produce reduced syntactic complexity)? Third, the multi-label activation model must be validated against cases with known comorbidity to determine whether the system accurately represents complex clinical presentations.

Fourth, cross-cultural and cross-linguistic validity must be established. The linguistic features predicted by NOSS are derived primarily from English-language research, and their validity in other languages and cultural contexts cannot be assumed. Fifth, test-retest reliability must be demonstrated: do the sector activation scores remain stable across repeated assessments of the same individual in a stable clinical state? Sixth, sensitivity to change must be established: do activation scores change appropriately in response to effective treatment?

An early-stage implementation of principles derived from NOST and NOSS has been deployed through a clinical pilot program involving licensed practitioners. While formal results are not yet available for peer-reviewed reporting, the preliminary clinical observations support the feasibility of language-based cognitive assessment across multiple NOSS sectors. Independent validation by researchers without commercial interest remains essential.

Limitations

Several significant limitations must be acknowledged. First, NOSS is a theoretically derived classification system. The sector boundaries, predicted signatures, and proposed cross-sector dynamics are based on integrating existing literature rather than on direct empirical observation of oscillatory–linguistic correspondences. The system's validity is therefore entirely dependent on the accuracy of NOST's theoretical claims.

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Second, the 12-sector system may not capture all clinically relevant cognitive states. Conditions such as eating disorders, sleep disorders, and somatic symptom conditions are not explicitly represented in the current taxonomy, though they may produce secondary activation across existing sectors. Third, the choice of 54 linguistic features and 12 sectors involves judgment calls that affect the system's sensitivity and specificity in ways that can only be determined empirically.

Fourth, the Captivity Trauma sector (S9), while theoretically motivated, addresses a clinical domain with limited existing research on linguistic markers, making its predicted signatures more speculative than those of better-studied sectors. Fifth, the author's commercial interest in applications of this classification system constitutes a potential bias that independent investigators should weigh when evaluating the system's claims.

Conclusion

The Neural Oscillatory Spectroscopy System provides a 12-sector classification taxonomy designed to operationalize NOST for computational cognitive assessment. Its design principles—multi-label continuous activation, oscillatory grounding, and clinical comprehensiveness—distinguish it from existing classification approaches by combining dimensional measurement with mechanistic neuroscientific grounding.

The system's value ultimately depends on empirical validation. If the predicted oscillatory–linguistic signatures are confirmed across the 12 sectors, NOSS would provide a comprehensive framework for language-based cognitive assessment spanning the major domains of clinical practice. If specific predictions are refuted, the falsified sectors can be revised or removed while preserving the overall classificatory structure. In either case, the system provides

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a structured set of testable propositions that advances the empirical investigation of language-based cognitive assessment.

The research program required to validate NOSS is substantial but tractable. The scientific community's participation in this program—through independent testing of sector-specific predictions, cross-cultural validation, and critical scrutiny of the system's assumptions—is essential to determining whether NOSS represents a genuine advance in cognitive classification.

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